Dheeraj Kumawat

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.69
Khandelwal Public Senior Secondary School, Jobner	Class XII (Board Of Secondary Education, Rajasthan)	2018	84.80%
Ganesh Bal Vidhya Mandir Senior Secondary School, Hirnoda	Class X (Board Of Secondary Education, Rajasthan)	2016	86.33%

WORK EXPERIENCE

Industrial Training - Coach Care Complex, North Western Railways, Jaipur

- Studied maintenance of ICF and LHB railroad coaches, including bogie assemblies, suspension systems and wheel axle configurations.
- Obtained exposure to critical safety systems, including twin-pipe air brakes, FIBA (air-spring failure detection), and smoke/heat fire suppression systems.
- Learnt wheel defect inspection standards (flat tire, deep flange, hollow tire) using a tire defect gauge and studied bio-toilet waste management systems.

POSITIONS OF RESPONSIBILITY

Placement Coordinator -- IIT Hyderabad

- Placement Coordinator for the TFE branch--primary point of contact with the Training & Placement Cell, leading outreach to prospective recruiting companies.

PROJECTS

Comparative CFD Study (ANSYS Fluent and OpenFOAM) of a 4×4 Ducted Axial Fan Array, Validated Against Experimental Data

- Simulated a 4x4 ducted axial fan-stator array in ANSYS Fluent and OpenFOAM, capturing rotor-stator flow interaction.
- Validated CFD results against experimental data, benchmarking accuracy between commercial and open-source solvers.
- Analyzed aerodynamic performance trends across the fan array to study interactions between adjacent fans and stators.

2D CFD Simulation of Laminar Flow over a Backward-Facing Step:

- Solved 2D incompressible Navier-Stokes equations using a **stream function-vorticity (-) formulation** using **Python code** on a **250×40 finite-difference grid**.
- Implemented boundary conditions and solved the sparse **Poisson system** for velocity, vorticity, and streamline fields.
- Generated **vorticity/velocity contours and streamline plots**, with quantitative velocity profiles at x = 25h and 50h capturing flow reattachment.

Supersonic Nozzle Design via Method of Characteristics and CFD Validation

- Designed a supersonic nozzle contour using **Method of Characteristics (MOC)** in MATLAB to achieve a target exit Mach number (Me = 2.157).
- Meshed (**14,141 cells**, skewness < 0.25) and simulated in **ANSYS Fluent** (density-based, inviscid, Roe-FDS).
- Validated design across MOC, isentropic theory, and CFD -- exit Mach agreement within **2.2%**.

Optimization of Design Parameters of Straight-Channel Printed Circuit Heat Exchanger for Supercritical CO Brayton Cycle-Based Concentrating Solar Power Plants

- To achieve **90%** heat transfer effectiveness across a **2-100 MW** power range, developed a **MATLAB**-based **optimization model** for PCHE design, benchmarking **6 nature-inspired optimization algorithms** (ABC, GWO, Cuckoo, JAYA, PSO, and TLBO).
- Validated the model against published research with only **0.07-2.4% error and** identified the **Cuckoo algorithm** as most efficient--delivering the smallest, most **material-efficient design** (0.26 m^3) in the fastest simulation time (74.6 s)
- Integrated optimization results with **CFD analysis** in **ANSYS Fluent**, enabling further design refinement and validation.
- Co-authored and published the study as a conference paper in *Advances in Clean Energy and Sustainability, Vol. 2* (Springer), DOI: https://doi.org/10.1007/978-981-97-5419-9_1.

SKILLS

Software: SolidWorks, AutoCAD, Ansys Workbench, Ansys Fluent, ICEM CFD, Ansys Turbogrid, OpenFOAM

Programming languages: Python, MATLAB

Other skills: Problem-solving and analytical skills, Communication skills

EXTRACURRICULAR

Engage in **Badminton**, **Cricket** for recreation and fitness.